

# Mohak Kapoor

🌐 Portfolio    ✉ contact.mohakapoor@gmail.com    ☎ +91 98702-32056    in Mohak Kapoor    🌐 mohakapoor

## Technical Skills

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| <b>Programming Languages</b>  | Python (Advanced), C++                                             |
| <b>DevOps / Backend</b>       | GitHub Actions, Docker, CI/CD, PostgreSQL, REST APIs               |
| <b>Machine Learning</b>       | Regression, Classification, Deep Learning, Time-Series Forecasting |
| <b>Libraries / Frameworks</b> | TensorFlow, scikit-learn, Pandas, NumPy, Django, Polars            |

## Projects

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### Cross-Market Index Forecasting Model (CMFM v1.0) Jan-May 2025

- Developed a deep learning model to predict index bid-open prices using 7M rows of 1-minute Dukascopy data, achieving MAPE of 3.1% (Japan), 4.2% (UK), and 12% (US) on 2025 test data.
- Built a scalable Python/Polars/TensorFlow pipeline, processing data in chunks, engineering features like midprice, MACD, lagged log returns, & z-score normalization and market session-aware null imputation.
- Implemented a CNN-LSTM model with self-attention, using masks for market closures and L1/L2 regularization, optimized for i9/32GB RAM with GPU memory management.

### CaptchaOCR — CAPTCHA Recognition System [↗](#)

- Built end-to-end CAPTCHA recognition system using CRNN architecture with CTC loss, achieving 96%+ character accuracy.
- Implemented CNN + BiLSTM model with synthetic data generation and GPU acceleration. Deployed on Hugging Face. Tools: Python, PyTorch, CNN-LSTM, Computer Vision.

### Solar Power Generation Predictor [↗](#)

- Developed a TensorFlow model to predict solar power output using OpenWeatherMap API, geographical, and time-based features, hosted on a Django web platform.
- Tools: Python, sklearn, TensorFlow, Pandas, NumPy, Django, OpenWeatherMap API

## Experience

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### JPMorgan Chase & Co. Quantitative Research Virtual Experience Jan 2025

- Analyzed a book of loans to estimate customer probability of default, applying quantitative research methods in a simulated environment.
- Used dynamic programming to convert FICO scores into categorical data, enhancing default prediction model robustness.

### HumanizeIQ — AI Intern Integrations June 2025 – Present

- Developed two specialized MCP servers using Cloudflare Workers: image generation and multimodal document generation (text + images), with async processing and R2 bucket storage.
- Built scalable MCP infrastructure implementing background processing, result retrieval, and integration with LLM chat systems.
- Designed AI-powered workflow orchestration for recruiter call analysis using Gemini and GPT-4 models, automating report generation and email distribution.
- Set up APIs for report storage, email distribution, and call event handling to support the recruiter analysis workflow.

## Education

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### Jaypee Institute of Information Technology, Noida-62 Sept 2022 – July 2026

*B.Tech in Electronics and Communication Engineering*

- Relevant Coursework:** Introduction to Computer Systems, Fundamentals of CS, Multivariate Calculus, Differential & Integral Calculus, ML and Deep Learning, Control Systems, A in Advanced Statistics